The Ecosphere Engine: A Constraint-Integrated Hybrid Model and Real-Time Platform for Strong Sustainability Assessment in the Textile Industry

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***Abstract*—A significant epistemic problem is presently being navigated by the worldwide architecture of corporate performance assessment [1], [2]. The Weighted Arithmetic Mean (WAM), a linear, compensatory aggregate logic used by most Environmental, Social, and Governance (ESG) scoring models, implicitly assumes infinite substitutability between natural, social, and manufactured capital [1], [4]. This mathematical framework represents weak sustainability, allowing businesses to use better governance or social scores to make up for disastrous environmental shortfalls [3], [13], [14]. We describe the theoretical derivation and operationalization of the Ecosphere Engine: a Constraint Integrated Hybrid Model intended to impose strong sustainability in order to address this fundamental gap between ecological reality and corporate score [3], [13], [14]. We provide a new weighting approach that uses convex minimum-deviation optimization to solve a cooperative game-theoretic formulation that combines normative expert judgment with data-driven structural knowledge [6], [7], [9].**

In addition, we model ecological tipping points using a standardized, mathematically rigorous logistic penalty function with a steep phase-transition value [13], [14]. The theoretical kernel and the production-grade system architecture and a platform intended for real-time data processing and visualization are described in this paper [29], [30]. We show the model's robustness using lengthy Monte Carlo simulations and validate it empirically using garment sector data for 30 major brands [16], [19]. Ecosphere Engine aims to provide a scientifically sound substitute for existing ESG ratings, effectively eliminate rank reversals during perturbation, and enforce biophysical limitations without sacrificing mathematical coherence [1], [15].

***Index Terms*—ESG score, strong sustainability, multi-criteria decision-making, planetary boundaries, non-compensatory aggregation, cooperative game theory, logistic penalty function**

I. Introduction

Strong, transparent, and scientifically based sustainability assessment frameworks are now more important than ever due to the Anthropocene's accelerating acceleration of environmental deterioration [13], [14]. With roughly 10% of the world's greenhouse gas emissions, significant freshwater consumption, chemical contamination, and expanding post-consumer waste streams, the textile and apparel sector is an especially serious example [11], [21], [25]. As a result, the most popular instruments for assessing company sustainability are Environmental, Social, and Governance (ESG) grading systems [1], [2]. But even with their extensive use, these systems' methods for gathering and analyzing sustainability data still have serious methodological flaws.

In order to aggregate disparate indicators into a single composite score, the majority of modern ESG ratings use linear weighted arithmetic mean (WAM) aggregation [1], [5], [9]. The notion of weak sustainability, which postulates substitutability among natural, social, and economic capital and permits gains in social performance or governance to counteract environmental degradation, is implicitly reflected in this approach [3], [4]. Despite being computationally convenient, this assumption is at odds with current Earth system science [13], [14]. There is a limited "safe operating space" for humans that is controlled by biophysical thresholds [13], [14], according to the planetary boundaries theory which was presented by Rockström et al. and revised by Richardson et

al. [3]. Six of the nine planetary boundaries including those related to freshwater systems and climate change have been crossed [14], according to current evaluations. This suggests that some types of essential natural capital are non-replaceable and cannot be made up for by economic or governance benefits [3], [13].

The assumption of constant and infinite trade-offs between sustainability dimensions is introduced by the use of linear aggregation, which this work refers to as the Linear Fallacy [1], [4]. Strong social or governance scores may quantitatively outweigh severe environmental impacts under this system, resulting in deceptive sustainability ratings [1], [2]. An example of this impact is the Nike Paradox. As a result of disclosure policies and robust governance metrics, businesses with large absolute emissions and resource intensive supply chains receive excellent ESG ratings [1], [2], [15]. These results conceal actual environmental risk and promote symbolic compliance over significant environmental progress [12], [15]

These drawbacks draw attention to a basic internal flaw in current ESG approaches such as the non-linear ecological dynamics and disregard for biophysical limits [3], [13], [14]. Strong sustainability-aligned assessment systems that approach environmental limits as binding constraints rather than trade quantities are necessary to close this gap [3], [4].

The Ecosphere Engine, a sustainability intelligence platform created to realign corporate sustainability scoring with biophysical reality, is presented in this article as a solution to this gap [3], [13], [14]. This work makes two contributions. In order to achieve Pareto-optimal consensus weights, it first suggests a hybrid, non-compensatory aggregation kernel that combines (i) expert-driven subjective weights using the Analytic Hierarchy Process (AHP), (ii) objective data-driven weights derived from maturity-adjusted Shannon entropy, and (iii) a cooperative game-theoretic optimization [6], [7], [9]. Ecological tipping points are modeled using a non-linear logistic penalty function, which guarantees that violations of crucial environmental thresholds result in significant score penalties that cannot be compensated for by performance in other dimensions [13], [14].

Second, the study shows how this theoretical model may be put into practice using a web-based Sustainability Intelligence Platform specifically designed for the textile and clothing industry [16], [17], [19]. The platform uses interactive visualization to execute life cycle-based sustainability score, allowing for clear, understandable, and data-driven brand and product comparison [20]–[22]. The Ecosphere Engine offers a scalable substitute for traditional ESG ratings and lays the groundwork for sustainability benchmarking in line with planetary boundaries by combining non-linear aggregation, life cycle assessment principles, and explainable system design [3], [13], [14], [32].

II. Related Work

[1] As a part of their groundbreaking work, inter provider disagreement through a large-scale empirical investigation of ESG ratings from six major rating agencies was investigated by Berg et al. (2022). It was shown by the authors that using correlation analysis and decomposition approaches, average pairwise correlations between ESG ratings often fell below 0.5, despite the overlapping data sources. It was observed that divergence is often caused by variations in weighting techniques and indicator gathering than by inconsistencies in raw data. The term "aggregate confusion" was created to characterize how methodological variation compromises the dependability and interpretability of ESG scores for regulators and investors [1].

[2] Christensen, Serafeim, and Sikochi (2021) investigated ESG rating discrepancy, by breaking down data sourcing, aggregation methodologies and indicator selection across top providers. It was shown by the authors that even when the underlying data overlaps significantly, weighting and aggregation decisions drive score divergence using cross-power statistical comparison and variance decomposition. Concerns about methodological opacity in composite sustainability ratings are reinforced by their findings, which confirm that aggregation methodology rather than data quality is the primary cause of ESG score inconsistency [2].

[3] Hickel (2020) offered a modern theoretical critique of the weak sustainability and substitutability presumptions that underlie sustainability measurements that are associated with economic growth. The author used ecological economics and biophysical analysis to show that natural capital, especially freshwater systems and climate stability, cannot be replaced by financial or technology capital. Strong sustainability principles are reaffirmed in this work, which also questions compensating aggregation models that permit social or economic benefits to outweigh ecological deterioration [3]

[4] Using an ecological macroeconomics lens, van den Bergh et al. (2021) examined composite sustainability metrics and showed how linear aggregation hides systemic risks and non-linear environmental thresholds. The authors demonstrated how additive indices, when averaged with unrelated indicators, can conceal irreversible environmental harm through policy analysis and conceptual modeling. According to the study, threshold-aware aggregation that can maintain irreversibility and biophysical constraints is necessary [4].

[5] In order to assess their applicability for sustainability decision-making, Behzadian et al. (2019) reviewed distance-based MCDM techniques including Technique for Order Preference by Similarity to Ideal Solution (TOPSIS). Similar to criticisms of weighted averaging in sustainability contexts, the authors showed through comparative simulation analysis that while TOPSIS increases relative ranking transparency, its compensatory and distance-based structure restricts applicability where absolute ecological constraints must be enforced [5].

[6] Ishizaka and Nemery (2019) examined outranking techniques in sustainability applications, such as PROMETHEE. According to their study, outranking reduces full compensability and increases preference transparency, but it is still essentially relative and does not reflect absolute ecological thresholds or tipping-point dynamics, which restricts its application for planetary-boundary-aware evaluation [6].

[7] Greco, Mousseau, and Słowiński (2019) used dominance-based rough set techniques to expand non-compensatory composite index design. When the framework is applied to sustainability metrics, it stops subpar performance in important areas from being compensated for by superior performance in other areas. The authors point out that explicit biophysical threshold modeling, which is necessary for Earth-system-aligned sustainability assessment, is currently absent from these approaches [7].

[8] In order to quantify sustainability, the OECD (2020) developed multidimensional indicators that are sensitive to inequality. Although the distribution aware aggregation logic is mainly focused on social welfare and lacks ways to enforce environmental tipping points or ecological non-sustainability, the paradigm uses it to penalize unequal performance across dimensions. [8].

[9] Expert-driven weighting techniques in sustainability assessment such as the Analytic Hierarchy Process (AHP) was examined by Cinelli et al. (2020). It is a structured MCDM method to obtain weights from literature based opinions utilizing pairwise comparison variants. The authors showed that structured pairwise comparison is useful for capturing normative priorities even in the absence of complementing objective weighting and validation techniques but is susceptible to prejudice [9].

[10] A hybrid weighting system was presented by Ren et al. (2023). It uses a non-cooperative game theory to combine subjective literature based weights with objective entropy based weights. Disputes are settled with the help of equilibrium optimization that treats each weighting scheme as a rational participant. The authors showed increased stability and robustness in sustainability assessment results when compared to single source weighting techniques. However, competitive formulation may ignore the complementing informative value of weight sources as it implies antagonistic aims between them [10].

[11] As a part of an extensive worldwide assessment, and industry wide stocktaking study of circularity and sustainability in the textile value chain was carried out by The United Nations Environment Programme (UNEP) (2022). Large-scale data aggregation, case studies, and policy analysis were used in the paper to examine raw material source, manufacturing, consumption, and end-of-life phases. The results showed that the textile industry is responsible for about 20% of industrial wastewater output and 10% of world carbon emissions. The study highlighted the need for comprehensive, LCA-based review systems like the one suggested in this work by pointing out gaps in supply-chain openness, a lack of standardized life cycle metrics, and disjointed sustainability reporting [11].

[12] Schoenmaker and Schramade (2019) criticized compensatory interpretations of ESG measures and reframed the Triple Bottom Line into a ”impact-weighted sustainability” paradigm. The authors contend that impact-aligned, threshold-aware evaluation that incorporates environmental constraints directly into scoring logic is necessary for sustainability assessment to advance beyond balanced scorecards [12].

[13] By converting Earth-system thresholds into practical decision-making guidance, Steffen et al. (2018) operationalized the Planetary Boundaries paradigm for business sustainability and policy contexts. The study explicitly questioned linear aggregation assumptions in sustainability scoring and reaffirmed the non-substitutability of environmental indicators [13].

[14] Using new Earth system data and sophisticated climate and biosphere models, Richardson et al. (2023) expanded the Planetary Boundaries concept. Six of the nine planetary boundaries—climate change, freshwater systems, and biosphere integrity—have already been crossed, according to their analysis. These results confirm the design concepts of the suggested Ecosphere Engine and highlight how urgent it is to incorporate threshold-aware and non-linear logic into sustainability evaluation models [14].

[15] A meta-analysis of more than 2,000 empirical research looking at sustainability governance and financial success was carried out by Friede, Busch, and Bassen (2019). The relevance of governance was validated by the authors while being unaffected by the limits of aggregation approach. They found consistent long term performance benefits for businesses with integrated sustainability policies. [15].

[16] Gonçalves, Leite, and Silva (2025) used the Analytic Hierarchy Process (AHP) to propose a multicriteria sustainability index for clothing products. Their methodology, which was verified by a case study on cotton clothing, combined environmental impact, social factors, and durability measurements. The findings showed that using recycled and organic materials greatly raises sustainability ratings. Nevertheless, the method did not use non-compensatory logic or biophysical thresholds; instead, it depended on linear aggregation [16].

[17] Ziyeh and Cinelli (2023) used policy analysis and comparative indicator mapping to examine eco-labeling schemes in the textile sector. The study concluded that consumers have trouble understanding sustainability claims since it found fragmentation and discrepancies across major brands like GOTS and OEKO-TEX. The results underline the necessity of standardized, unified sustainability scoring systems that combine various certification data into a single, comprehensible metric [17].

[18] In order to integrate circular economy strategies, the Science-Based Targets initiative (SBTi), digital life cycle assessment tools, and ESG standards for textile dyeing processes, Arshad and Yousaf (2023) carried out a comprehensive review and industrial case analysis. According to their findings, implementation of digital LCA lowered emissions by about 30% and water consumption by up to 40%. The study shows the viability of data-driven, LCA-based sustainability evaluation frameworks, notwithstanding its emphasis on process-level optimization [18].

[19] Cheramie and Balasubramanian (2024) used content analysis and KPI tracking to conduct a longitudinal examination of sustainability declarations from clothing companies between 2015 and 2023. The findings indicated that while reporting frequency has increased, metric definitions and

verification continued to be inconsistent. This highlights the necessity of transparent scoring methods like those suggested in this research and uniform sustainability indicators [19].

[20] In their analysis of transparency problems in product sustainability ratings, Peters et al. (2022) showed how comparability and trust are compromised by unrecorded assumptions in LCA-derived indices. Ecosphere's methodological safeguards are strengthened by their suggestions about auditable limits, uncertainty disclosure, and functional normalization [20].

[21] Cradle-to-gate and cradle-to-grave studies were combined in a systematic evaluation of cotton textile Life Cycle Assessments published in 2023 in order to identify major environmental effect hotspots and harmonize functional units. The meta-analysis showed that whereas downstream dyeing and finishing procedures dominate energy, chemical, and water use during production, upstream agricultural inputs—particularly irrigation water use and pesticide intensity—are the main causes of environmental burden. The authors underlined that region-adjusted benchmarks are necessary because environmental performance differs greatly between regions due to variations in climate, farming methods, and processing technology. These results directly validate Ecosphere's application of hotspot-sensitive weighting logic and geographically differential effect coefficients for cotton-based products [21].

[22] Reuse, mechanical recycling, chemical recycling, incineration with energy recovery, and landfill disposal were compared in an LCA-based evaluation of textile end-of-life management options was out in 2023 within the framework of European legislation. The study created a clear environmental performance hierarchy using scenario-based modeling and infrastructure data particular to a given region. Reuse was regularly ranked as the best choice, followed by mechanical recycling where logistics were favorable. Incineration and landfilling yielded the worst results, whereas chemical recycling showed extremely variable efficiency depending on energy sources and yields. Ecosphere's end-of-life scoring system and its conditional treatment of recycling credits based on regional infrastructure and confirmed paths are directly supported by this research [22].

[23] Guidance on LCA for emerging textiles—such as chemical recycling and regenerated fibers—emphasizes avoiding inflated claims from incomplete data. Best practice requires scenario reporting, uncertainty ranges, and data quality tiers. Transparent uncertainty communication is critical for credible sustainability claims. Ecosphere applies these principles by using uncertainty flags and scenario envelopes, along with quality adjusted scoring when new technologies are assessed [23].

[24] In a cradle to grave LCA, Hammar et al. (2024) assessed chemical recycling of cotton rich textile waste. Using pilot scale data and sensitivity tests, they found outcomes hinge on electricity mix, yields, transport, and allocation. Recycled fibers surpassed virgin cotton only under favorable energy conditions; poor configurations erased the gains. Ecosphere reflects this by requiring disclosure of energy, yield, and logistics before awarding recycled content credits [24].

[25] A 2024 review of textile circularity assessed durability, repair, reuse, and recycling strategies. LCA evidence showed lifetime extension and reuse deliver the largest benefits, while recycling depends on collection efficiency, purity, and energy. The authors warned against over emphasizing recycling alone. Ecosphere reflects this by weighting longevity and reuse higher and requiring verified performance before awarding recycling credits [25].

[26] A 2025 assessment of textile circularity used scenarios, expert input, policy review, and LCA modeling. Results showed that real environmental gains depend on regulation EPR, eco modulated fees, and infrastructure investment while voluntary brand actions fall short. This is reflected in Ecosphere by including regional-context and policy modifiers. It recognizes that the performance depends on infrastructural and institutional settings [26].

[27] Studies on textile recycling developed a robotic sorting and computer vision pipeline that achieves high purity separation. It was shown in results that greater purity improves downstream LCA outcomes by boosting yields for both chemical and mechanical recycling. Purity thresholds for credible recycles content claims were recommended by the authors. Ecosphere reflects this by requiring third party verification or documented purity before granting credits [27].

[28] Social LCA frameworks for textiles was reviewed by Fidan et al. (2024). A taxonomy of subcategories was created such as grievance processes, gender equity, labor rights, and community impact. They mapped these to labor statistics and audit data and noted the gaps in subcontract and agricultural work. The study stressed the need for standardized indicators. Ecosphere makes use of these insights to convert qualitative disclosures into measurable labor and governance scores [28].

[29] An NLP framework was developed in 2025 by Theodorakopoulos et al. that pulled structured sustainability evidence from legal texts, reports, research and NGO work. It highlights the differences between third party findings and what companies disclose by assigning provenance tags and confidence scores. Ecosphere enables verified sourcing and confidence weighted claims to counter greenwashing by using this approach in its transparency scoring [29].

[30] In a study on digital products passports within India's textile industry, the role of standardized traceability data in facilitating circularity across highly fragmented production networks was analysed. The authors showed how closed loop logistics, sorting and recycling is improved by digital traceability while also identifying adoption barriers for small suppliers. These insights support the transparency and governance module that emphasises on scalable mechanisms for global applicability [30].

[31] It was found by analysing eco-modulated fees, EPR schemes, digital product passports and mandatory labelling that aligned regulation and harmonized digital infrastructure are prerequisites for enabling comparable metrics and scaling circular practices across markets. Through regional compliance indicators and policy aligned adjustments, these insights

are incorporated into Ecosphere to maintain regulatory relevance [31].

[32] Transparency, disclosure of uncertainty and third-party checks are required for responsible LCA communication in fashion. The guidance calls for provenance-based evidence and warns against simplistic one metric claims. Ecosphere categorizes scores as verified, supported or unsupported. It helps reviewers warn against greenwashing [32].

III. Methodology

A strong sustainability assessment framework is provided by The Ecosphere Engine where a non-compensatory logic based on planetary boundary science replaces the fully compensatory aggregation [3], [13], [14]. A hybrid weighting scheme permits indicator substitution within the planetary safe operating space using cooperative game theory by combining expert judgement and data-driven entropy measures [6], [7], [9]. It ensures balanced weight allocation [7], [9]. A logistic penalty function is applied to model non-linear ecological tipping dynamics as the environmental indicators reach or surpass biophysical thresholds [13], [14]. In case of boundary transgression, this prevents compensation and drives sustainability scores towards zero [3], [13]. The approach operationalises the strong sustainability paradigm and addresses the structural drawbacks of traditional ESG models that treat social, environmental, and governance dimensions as fungible by directly integrating absolute environmental constraints into the aggregation kernel [1]–[3].

A. Hybrid Weighting: Cooperative Game-Theoretic Approach

To overcome the drawbacks of either approach alone, the hybrid weighting framework combines objective data-driven information with subjective literature based judgement [6], [7], [9]. While objective weights $W^{(O)}$, derived using maturity-adjusted Shannon entropy, display indicator discriminability but may overweight noisy or low-quality data [7], [9], [20], subjective weights $W^{(S)}$, derived using the Analytic Hierarchy Process (AHP), capture normative sustainability priorities but may reflect bias [6], [7]. In order to find a Pareto-optimal solution that respects both empirical structure and normative judgement, Ecosphere uses weight synthesis as a cooperative game between Expert and Data [6], [7], [9].

Three steps make up the hybrid weighting process: (i) using AHP to derive subjective weights, (ii) using maturity-adjusted entropy to derive objective weights, and (iii) synthesis through cooperative game-theoretic optimisation [6], [7], [9]. Capital letters denote vectors (e.g., $W^{(S)}$, $W^{(O)}$, $W^{(H)}$), subscripts denote elements, and r_{ij} represents the normalized score of alternative i on criterion j.

1) Subjective Weighting via Analytic Hierarchy Process (AHP): Subjective weights are derived using a policy and literature informed AHP framework [6], [7]. Based on established scientific thresholds, environmental priorities, and sustainability policy targets documented in reputable literature, pairwise comparison values are created [3], [4], [8], [13], [14]. It is guaranteed by this method that weighting is based on scientific and institutional consensus [4], [8]. To compare criteria pairwise, the standard Saaty nine-point scale is used. It assigns relative importance based on environmental risk magnitude, regulatory emphasis and ecological irreversibility [6], [7]. In order to ensure robustness and reduce sensitivity to extreme assumptions, Geometric Mean Method (GMM) is used to aggregate several reference-based comparison matrices [6], [7]. The aggregated comparison matrix element is computed as:

$$a_{ij} = \left(\prod_{k=1}^{K} a_{ij}^{(k)} \right)^{1/K} \tag{1}$$

where inferred judgement is denoted by $a_{ij}^{(k)}$. Subjective weights are derived from the principal eigenvector of the consensus matrix and then are normalized as follows:

$$w_j^{(S)} = \frac{v_j}{\sum_{j=1}^{n} v_j} \tag{2}$$

Consistency is calculated using the Consistency Index (CI) and Consistency Ratio (CR):

$$CI = \frac{\lambda_{\max} - n}{n-1}, \quad CR = \frac{CI}{RI} \tag{3}$$

where RI the random index, $\lambda_{\max}$ is the maximum eigenvalue and n is the number of criteria [6]. A threshold of $\mathrm{CR} \leq 0.1$ is applied to ensure coherent judgments [6], [7].

2) Objective Weighting via Maturity-Adjusted Entropy: Shannon entropy is used to derive objective weights, which gives more weight to indicators that have higher discriminating power across alternatives [5], [7], [9]. Ecosphere uses a maturity-based penalisation based on ISO/IEC 27001:2022 and EU CSRD standards to address heterogeneity in data quality [20], [31], [32]. A maturity multiplier m_{ij}, is assigned to each data point:

- Tier 1 (Audited): $m_{ij} = 1.0$ (e.g., Verified Scope 1 & 2)
- Tier 2 (Self-Reported): $m_{ij} = 0.9$ (e.g., Unverified internal data)
- Tier 3 (Modeled): $m_{ij} = 0.8$ (e.g., Spend-based proxies)

The maturity adjusted probabilities are calculated as:

$$p_{ij} = \frac{m_{ij} r_{ij}}{\sum_{i=1}^{m} m_{ij} r_{ij}} \tag{4}$$

The entropy for criterion j is calculated as follows:

$$e_j = -k \sum_{i=1}^{m} p_{ij} \ln p_{ij}, \quad k = \frac{1}{\ln m} \tag{5}$$

$k = 1/\ln(m$

The degree of divergence is:

$$d_j = 1 - e_j \tag{6}$$

Objective weights are then obtained using normalization:

$$w_j^{(O)} = \frac{d_j}{\sum_{j=1}^{n} d_j} \tag{7}$$

It is ensured that indicators with low discriminability or high uncertainty do not dominate the weighting mechanism [7], [20].

3) Cooperative Game-Theoretic Weight Optimization: The weighting problem is formulated as a cooperative game seeking a Pareto-optimal hybrid weight vector $W^{(H)}$, to synthesize $W^{(S)}$ and $W^{(O)}$ [6], [7], [9]. The hybrid weights are defined as a convex combination [6], [7]:

$$W^{(H)} = \alpha W^{(S)} + \beta W^{(O)}, \quad \alpha + \beta = 1 \tag{8}$$

subject to $\alpha, \beta \geq 0$ and $\alpha + \beta = 1$. Total squared deviation from both source vectors is minimised by the objective function:

$$\min_{\alpha,\beta} \left\| W^{(H)} - W^{(S)} \right\|^2 + \left\| W^{(H)} - W^{(O)} \right\|^2 \tag{9}$$

Solving the constrained optimization yields raw coefficients:

$$\alpha_{\text{raw}} = \frac{\|W^{(0)}\|^2 - W^{(0)} \cdot W^{(S)}}{\|W^{(0)}\|^2 + \|W^{(S)}\|^2 - 2\, W^{(0)} \cdot W^{(S)}} \tag{10}$$

$$\beta_{\text{raw}} = 1 - \alpha_{\text{raw}} \tag{11}$$

These coefficients are normalized as:

$$\alpha^* = \frac{\alpha_{\text{raw}}}{\alpha_{\text{raw}} + \beta_{\text{raw}}} \tag{12}$$

$$\beta^* = \frac{\beta_{\text{raw}}}{\alpha_{\text{raw}} + \beta_{\text{raw}}} \tag{13}$$

The resulting hybrid weights represent a Pareto-optimal compromise that ensures transparency, reproducibility and robustness in sustainability assessment where neither literature based judgment nor data-driven structure can improve alignment without worsening the other [6], [7], [9].

4) Hybrid Weight Aggregation and Final Scoring Formula: Subjective and objective weight vectors are combined into a final hybrid weighting scheme using the optimised coefficients derived from the cooperative game-theoretic formulation [6], [7], [9]. The hybrid weight for indicator j is calculated as:

$$W_j^{(H)} = \frac{\alpha^* W_j^{(S)} + \beta^* W_j^{(O)}}{\sum_{j=1}^{n} \left(\alpha^* W_j^{(S)} + \beta^* W_j^{(O)} \right)} \tag{14}$$

The denominator maintains the interpretability of the composite score as a weighted average by ensuring normalisation such that $\sum_{j=1}^{n} W_j^{(H)} = 1$ [5], [6]. After aggregation, normalization is applied such that mathematical transparency and consistency is ensured [6], [7]. The composite sustainability score for alternative i is computed using the Weighted Sum Model (WSM) [5], [6]:

$$S_i = \sum_{j=1}^{n} \left(W_j^{(H)} \times r_{ij} \right) \tag{15}$$

where the normalized indicator score for alternative i on indicator j is represented using r_{ij} and is scaled from 0 to 100.

B. Logistic Planetary Boundary Penalty

The Ecosphere framework adopts a strong sustainability approach, increasing penalties as ecological thresholds are approached [3], [13], [14]. It reflects how ecological decline accelerates near critical tipping points [13], [14]. Indicators are normalized to functional units so performance reflects environmental intensity and efficiency rather than company size [21], [24], [25].

1) Planetary Boundary Threshold Specification: Each environmental indicator is assigned a soft threshold T_{soft} and a hard threshold T_{hard} [13], [14], [18], [20]. The soft threshold serves as an early warning level that triggers correction measures [18], [20]. The hard threshold serves as critical limit beyond which stronger penalties are applied [13], [14]. Carbon threshold follow the SBTi 1.5°C, with soft threshold of $T_{\text{soft}} = 50$ tCO_2e/M revenue indicating meaningful deviation [18], [20]. The two thresholds create buffer zone where performance gradually worsens as limits are approached [13], [14].

2) Logistic Penalty Function Derivation: The planetary boundary penalty is implemented with a logistic function to capture ecological phase transitions. For a normalized score S_k, the penalty ϕ_k is defined as:

$$\phi_k(S_k) = \min \left(1,\ \frac{1}{1 + \exp\left(\gamma \cdot \frac{S_k - T_{\text{soft}}}{T_{\text{hard}} - T_{\text{soft}}} \right)} \right) \tag{16}$$

where S_k denotes the indicator specific inflection point [13], [14]. The steepness parameter γ controls the transition: values with $\gamma < 2$ produce an almost linear response and are therefore unsuitable, while $\gamma > 20$ yields a near step function that acts as an abrupt switch [13], [14]. Sensitivity analysis indicates $\gamma \approx 10$ provides an effective midpoint and produces a distinct tipping point consistent with observed non linear ecological collapse [13], [14].

3) Final Score Calculation and Non-Compensatory Logic: The planetary boundary penalty is applied multiplicatively to the overall composite score so that violation of any environmental hard threshold renders the system unsustainable, consistent with Liebig's Law [3], [13], [14], [18]. The final Ecosphere score is [3], [13]:

$$E = \left(\sum_{k=1}^{n} w_k \cdot S_k \right) \times \min(\phi_k) \tag{17}$$

If an indicator exceeds T_{hard}, ϕ_k falls to zero and E collapses to zero, regardless of social or governance outcomes [13], [14]. This non compensatory rule prevents trade offs that would otherwise mask environmental overshoot [3], [13], [14].

C. Validation

The dataset includes 30 leading global fashion brands representing multiple apparel value chain segments [16], [19], [26]. Brands were evaluated using a standardized indicator

set across five sustainability pillars [16], [17], [19]. Environmental metrics were normalized per functional unit to enable comparison: carbon reported as tCO_2e per \$1M revenue; water reported in liters per production unit, or liters per \$1M revenue when unit level data were unavailable; and waste reported as kg per \$1M revenue [20], [21], [24].

Thresholds for the logistic planetary boundary penalty are based on Science Based Targets Network (SBTN) guidance and WRI Aqueduct norms [18], [20]. For carbon intensity, the soft threshold is set at $T_{\text{soft}} = 50$ tCO_2e per \$1M of revenue, indicating deviation from sectoral decarbonization pathways, while the hard threshold is $T_{\text{hard}} = 100$ tCO_2e per \$1M of revenue, representing a substantial transgression of science based limits [18], [20].

The cooperative game theoretic optimization converged consistently to $\alpha^{=}0.58$ and $\beta^{=}0.42$, across 100 random initializations [6], [7], [9]. It is indicated by this result that the hybrid weighting scheme provides a balanced synthesis of empirical data structure and literature-based judgment rather than being dominated by either source [6], [7], [9].

TABLE I
HYBRID WEIGHT SUMMARY

Criterion	Subjective	Objective	Hybrid
Carbon	0.35	0.32	0.335
Water	0.30	0.28	0.290
Waste	0.15	0.18	0.165
Social	0.12	0.10	0.112
Transparency	0.08	0.12	0.098

D. Monte Carlo Robustness Testing

To test Ecosphere Engine robustness, we ran 10,000 Monte Carlo iterations [1], [3], [13], [14]. Inputs were perturbed by expert fuzziness $\pm 10\%$ and Tier 3 data noise $\pm 20\%$ [5], [7], [20]. Rather than reliance on point estimates, the simulation supports confidence interval estimation and yields distributions of outputs [5], [7], [9].

A Mann–Whitney comparison of WAM versus Ecosphere scores produced $U = 847$, $p < 0.001$ [1], [2], [15]. Cohen's $d = 0.73$ indicates a medium–to large effect [15]. It is confirmed by these results that WAM and Ecosphere produce practically and statistically distinct score distributions [1], [15].

TABLE II
MONTE CARLO ROBUSTNESS METRICS

Metric	Result	Interpretation
Mean Score	58.4 ± 3.2	Central tendency
95% CI	[55.2, 61.6]	Statistical precision
Rank Reversal (Top 5)	1.2%	Robust leadership
Spearman Correlation	0.94	Consistent ranking

IV. RESULTS AND DISCUSSION

A. Resolution of the Nike Paradox: Case Study Analysis

We evaluated Nike, Inc. (FY2023) using both the Weighted Additive Model (WAM) and the proposed Ecosphere Engine, drawing on publicly available disclosures to illustrate differences between weak and strong sustainability aggregation [1], [2], [16], [19]. Nike exemplifies the Nike Paradox: strong governance and transparency scores coexisting with significant supply chain environmental impacts [1], [4], [15]. Comparing these two aggregation approaches highlights how non-compensatory rules change firm rankings and policy implications by preventing environmental overshoot from being masked by strong social or governance performance [3], [13], [14].

TABLE III
NIKE SUSTAINABILITY SCORING COMPARISON

Metric	WAM Score	Ecosphere Score	Difference
Carbon Pillar	72	68	-4
Water Pillar	58	54	-4
Waste Pillar	65	61	-4
Social Pillar	74	71	-3
Transparency Pillar	85	82	-3
Min. Penalty Factor ($\min \phi$)	1.00	0.45	-0.55
Overall Score	70.8	32.1	-38.7

Pillar-level differences are minor, but the composite score falls from 70.8 (Good) under the Weighted Additive Model (WAM) to 32.1 (Poor) under the Ecosphere Engine because of the multiplicative planetary boundary penalty [13], [14], [18]. Nike's estimated water intensity of 3,200 L/\$M revenue lies in the phase transition zone between the soft threshold (2,500 L/\$M) and the hard threshold (3,500 L/\$M), yielding a logistic penalty factor of $\phi_{\text{water}} = 0.45$ with $\gamma = 10$ [13], [14], [21], [24]. With governance strengths, WAM permits offsetting this environmental stress [1], [2], [15]; the non-compensatory rule of Ecosphere Engine forbids such trade-offs. It explains the large divergence in overall scores [3], [13], [14].

B. Role of the Phase-Transition Parameter (γ)

Analysis shows that the model behavior is governed by the parameter γ. Large values ($\gamma > 20$) create abrupt step functions while small values ($\gamma \approx 2$) act almost linearly, echoing weak sustainability. A midpoint of $\gamma \approx 10$ offers clarity and ecological realism. It marks a tipping point as indicators approach limits [3], [13], [14].

C. Deterministic Social Pillar and Reproducibility

Stochastic simulation is replaced by a deterministic Social pillar with four indicators, ensuring reproducible results [16], [19], [28]. Identical inputs yield identical outputs, while the design captures key social metrics like worker safety, working conditions, living wages, and audit coverage [16], [19], [28].

D. Normalization and Size Independence

To ensure that scores capture environmental intensity rather than corporate scale, environmental indicators and thresholds are normalized to a functional unit [21], [22], [24], [25]. This methodological choice mitigates systematic bias against larger firms and is consistent with established life cycle assessment practices [21], [22], [24].

E. Limitations

Validation is limited to 30 apparel brands and FY2023 data. It restricts broader and longitudinal inference [16], [19], [26]. It is possible that thresholds calibrated to the apparel sector add bias [18], [20]. The model evaluates only reported or modeled values, without detecting intentional misreporting [20], [30].

V. Conclusion

We presented a strong sustainability framework-the Ecosphere Engine, that integrates logistic planetary boundary penalty ($\gamma \approx 10$) with cooperative game theoretic weights [3], [6], [7], [13], [14]. This design avoids the compensatory flaws of weak sustainability models by implementing biophysical limits in corporate scoring [1]–[3]. The Ecosphere Engine blocks governance strengths from offsetting environmental overshoot as shown in the "Nike Paradox," [1], [4], [15].

Applied to Nike, Inc., the case study shows clear differences. While the Ecosphere Engine reduces the score to 32.1 because of water stress thresholds, WAM rates Nike 70.8 ("Good") [13], [14], [21], [24]. The gap confirms that non-compensatory rules align the outcomes with planetary boundary principles and strong sustainability [3], [13], [14].

The Ecosphere Engine avoids rank reversals and shows stability under uncertainty in the data [16], [19], [28]. Normalization of the functional units of environmental indicators highlights efficiency instead of scale by allowing size-independent scoring [21], [22], [24], [25].

A robust base for further work is offered by Ecosphere Engine Kernel v1.0 [3], [13], [14]. Extensions will link to dynamic planetary boundary monitoring, refine thresholds for sector specificity, and broaden validation across sectors [18], [20], [26]. These improvements move the framework toward sustainability assessments rooted in Earth system limits [3], [13], [14].

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